

GTR9 Pro Post-Boot Log Analysis

Beelink GTR9 Pro · CachyOS · Linux 7.2.8-2-cachyos · ry-verify 7.219.0
Captures of 2026-10-02, 18:20 PDT · print edition, revision 32

VERDICT

Healthy — no CRIT, HIGH, or MED findings.
Open: one LOW finding (L-3, identifiers in the new captures) and 20 INFO findings that need no action. Closed: five LOW findings, confirmed by these captures. ry-verify reports 299 OK, 0 FAIL, 0 WARN, and 0 GEN_FAIL; the fix-script --verify run exits 0; there is no taint, oops, or splat.

OPEN							CLOSED		
0	0	0	1	20	2	7	5	5	1
CRIT	HIGH	MED	LOW	INFO	WATCH	UNKNOWNNS	LOW	UNKNOWNNS	EVENTS

Open items: Sections 4–9 and 11. Closed items: Section 10. Watch and by-design items are not counted as findings.

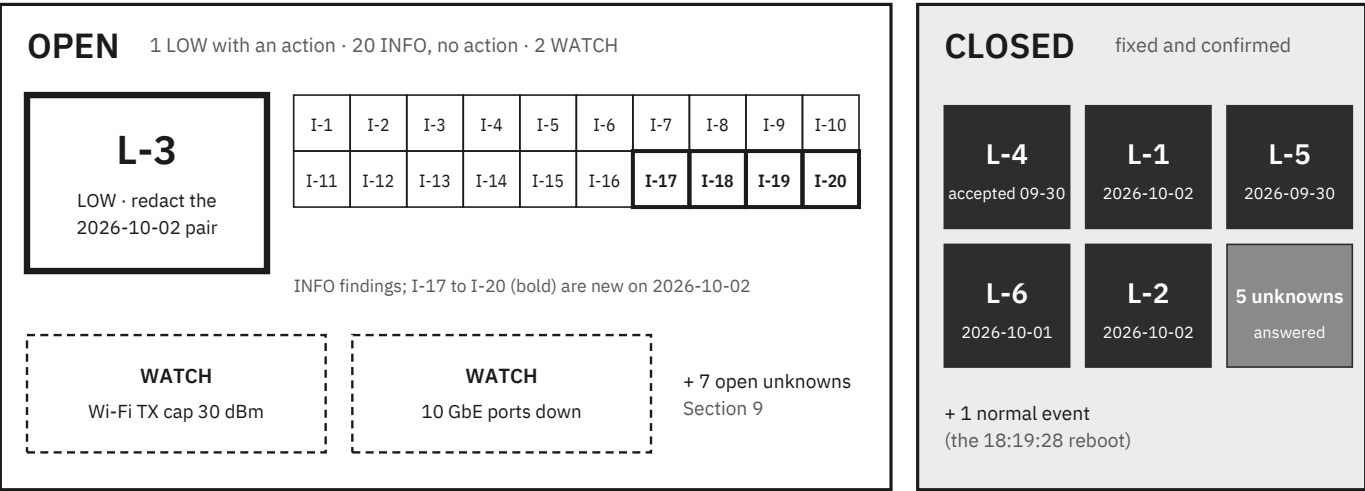


Figure 1 Open and closed items, 2026-10-02. Left: the open LOW finding, the INFO findings, the watch items, and the open unknowns. Right: closed LOW findings with their closing dates, the answered unknowns, and the normal reboot.

RY-VERIFY 7.219.0 299 OK 0 FAIL · 0 WARN · 0 GEN_FAIL	BOOT TO ROOT MOUNT 7.22 s +1.00 s against 2026-09-27	FAILED UNITS 0 / 0 system / user	KERNEL RING Clean no taint, oops, or splat	NVME (P310 2 TB) PASSED 42.9 °C · 8.92 TB written	GPU FIRMWARE MES 0x92 SMU up · DMUB 0x09005300
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Document control

Report	GTR9 Pro Post-Boot Log Analysis (2026-10-02), print edition
Revision	32, issued 2026-10-04
Captures	cachyos-bugreport.log, ry-verify 7.219.0 JSONL, gtr9-postboot-fix 2.3.0 --verify log; 2026-10-02 18:20 PDT
Masking	Identifiers in quoted lines are masked (L-3)
This revision	Editorial cleanup and consistent typefaces in tables, cards, and the contents; findings, values, and evidence unchanged since revision 30.

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1 Executive summary

The captures were taken after the 18:19 reboot on 2026-10-02, with every fix from revisions 1–19 in place. The 2026-09-27 captures are no longer on the host; Table 1 keeps their values.

Open: L-3, because 65 lines of the new captures carry identifiers; 20 INFO findings that need no action; two watch items; and seven unknowns. Closed: L-1, L-2, L-4, L-5, and L-6, five unknowns, and one event.

1.1 Key facts

- All five fixes hold after a reboot: no NetworkManager P2P warning (L-6), JACK through pw-jack with no JACK server (L-2), the soft mixer on the POROSVOC card (L-5), the SDL ignore list in the session (L-1), and only the accepted KWallet portal failure (L-4).
- The L-2 upgrade moved the kernel from 7.2.8-1 to 7.2.8-2-cachyos (clang 23.1.1) and Mesa from 26.2.3 to 26.2.4; the ry-install profile still matches (17/17 checksums, 15/15 kernel tokens live).
- Firmware: CPU microcode 0x0B700037 (early update), SMU initialized, DMUB 0x09005300, MES 0x00000092, and MT7925 Wi-Fi and Bluetooth firmware built 2026-08-13.
- The previous boot ran 2 h 10 min and adds I-17 to I-20: a wireless-extensions warning, a window that hung on close, a Bluetooth audio device connecting and disconnecting, and Steam's driver probe in the task manager (Figure 9).
- Idle captures: ry-verify at ≈ 39 s, the fix script at ≈ 49 s, and the bug report at ≈ 55 s after kernel start.

1.2 What changed since 2026-09-27

Revision 19 values come from the 2026-09-27 captures. The early boot runs about 1 s later; 0.8 s of that is the gap before the root mount (forced fsck, inferred; Section 9).

Table 1 Before and after

Item	2026-09-27 (revision 19)	2026-10-02	Change
Kernel	7.2.8-1-cachyos (clang 22.1.8)	7.2.8-2-cachyos (clang 23.1.1)	L-2 upgrade
Mesa	26.2.3	26.2.4	L-2 upgrade
ry-verify 7.219.0	299 OK; 0 FAIL, 0 WARN, 0 GEN_FAIL	299 OK; every section count equal	none
JACK provider	jack2 1.9.22 (jackd found by inxi)	pipewire-jack (pw-jack plugin)	L-2 closed
NM P2P warning per boot	1	0 in both boots	L-6 closed
D-Bus user-unit failures	1 current, 2 previous (names redacted)	1 current (KWallet portal), 2 previous	accepted, L-4 closed
Journal entries, current boot	55	53	L-6 warning and I-11 framebuffer line gone
Journal entries, previous boot	87 (2 min 42 s boot)	114 (2 h 10 min boot)	longer session
Root mounted	6.22 s	7.22 s	+1.00 s
No log output before the root mount	3.06 s	3.87 s	+0.81 s, fsck inferred
Wi-Fi associated	13.34 s	14.30 s	+0.96 s
Greeter / session hand-over	≈ 10.4 / ≈ 17.4 s	≈ 10.45 / ≈ 17.45 s	none
MES firmware	not logged; 0x91 read on 2026-08-14	0x00000092 (FL 2)	unknown closed
Failed units	not captured	0 system, 0 user (FL 2)	unknown closed
NVMe (Crucial P310 2 TB)	39.9 °C; 8.52 TB written; 119 days	42.9 °C; 8.92 TB; 124 days	+0.40 TB in 5 days
VRAM / GTT	32,768 / 48,091 MiB	32,768 / 48,091 MiB	none

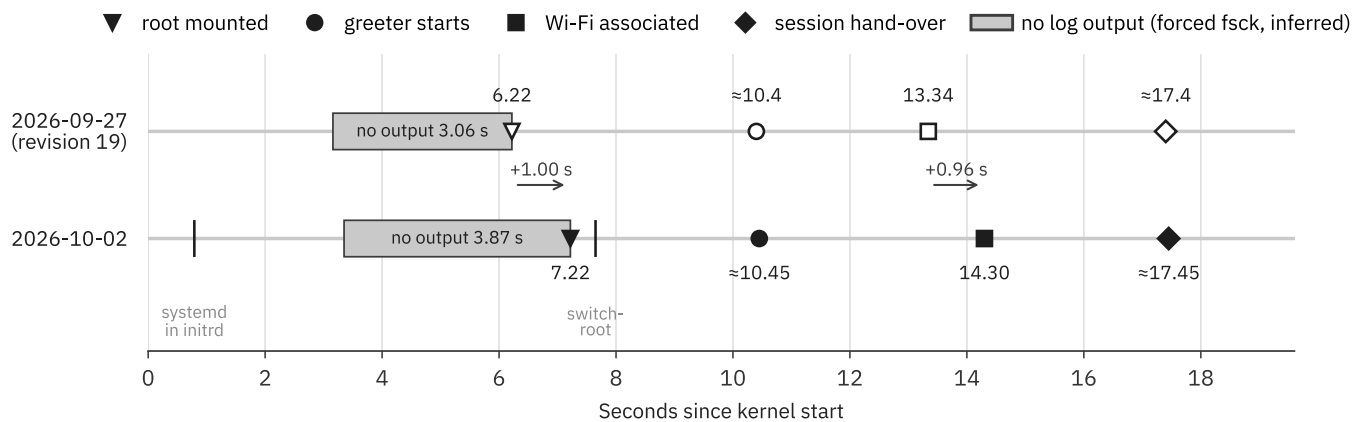


Figure 2 Boot milestones, 2026-09-27 against 2026-10-02 (Table 1). The root mount and Wi-Fi association move about 1 s later; the shaded box is the stretch without log output before the root mount, inferred to be the forced fsck. Hollow markers: 2026-09-27; filled: 2026-10-02.

1.3 Open actions

No INFO or watch item needs action. Table 2 lists the three open actions; Section 11 has their commands and a checklist.

Table 2 Open actions

#	Action	Why	How	Page
O-1	Redact the 2026-10-02 captures before posting them (L-3)	28 bug-report and 37 JSONL lines carry identifiers	Redact by hand; check with Section 11.1	p. 25
O-2	Merge the seven .pacnew files listed by the 2026-10-02 upgrade	House rule after every -Syu; skip if done since	<code>sudo pacdiff</code>	p. 25
O-3	Hand L-1, L-2, L-5, and L-6 to ry-install 7.224.0	The profile now carries all four; the old drop-ins duplicate them	Deploy, delete two drop-ins, run ry-verify	p. 25

2 Scope and method

2.1 Inputs

Table 3 Inputs

File	Bytes	Lines	Content
cachyos-bugreport.log	147,662	1,885	inxi, sched-ext state, full dmesg, journal of the current and previous boot (warning and above), package list
SHA256	5ee35ae048c94a328a49f45465d35ff3b2b587ab8d1aed335bde4dc3523dea92		
verify-20261002-182017-0700-1942.jsonl	51,436	483	ry-verify 7.219.0 static and runtime passes, profile gtr9_pro
SHA256	18d25f823a0778419e5026c4f37d61b895110b74d267ba34ec2f714389b82b41		
20261002-182027-3922.log	1,048	11	gtr9-postboot-fix 2.3.0 --verify: one record per step, exit status
SHA256	2160b2adc2d0f6077630d8f8e8f757bd0f0402a7eeca3abc3200432678f99440		

2.2 Capture window

Table 4 Capture window

Event	Wall clock (PDT)	Since kernel start
Previous boot (journal span)	16:08:42 → 18:19:29	ended by reboot at 18:19:28
Current boot: kernel start	≈18:19:38.05 (±0.16 s)	0 s
Greeter starts	18:19:48	≈10.45 s
Greeter → user session hand-over	18:19:55	≈17.45 s
ry-verify 7.219.0 run	18:20:17.118 → 18:20:19.451	≈39.06 → 41.40 s
Fix script 2.3.0 --verify	18:20:27.126 → 18:20:27.294	≈49.07 → 49.24 s
cachyos-bugreport capture	18:20:33	≈55.45 s

Kernel start is the intersection of the 1 s journal stamps of seven kernel messages with their dmesg offsets. Journal and bug-report times have 1 s resolution; offsets use the middle of the second (±0.5 s).

2.3 Method

- Every JSONL record, every journal entry of both boots, and all 1,359 dmesg lines were read and classified. The 124 lines matching the failure keyword set (Table 5) and the 205 dmesg lines matching a wider notice pattern are each attributed (Section 8).
- New messages were traced to their emitting code on 2026-10-02 (Linux v7.2, BlueZ, KWin, plasma-desktop, pulseaudio-qt, Chromium, CachyOS-Settings); unchanged messages keep revision 19's 2026-09-29 checks. Quoted lines are verbatim apart from masked identifiers and dropped date and host prefixes; each cites its line (BR, VJ, FL: Appendix B).
- Severity follows the observed impact on this host, not the log level. Commands are fish-compatible, one per line, and never wrapped.

Table 5 Failure keyword set (38 terms, case-insensitive, whole words)

Terms				
fail	failed	failure	error	warn
warning	unable	cannot	can't	could not
couldn't	not supported	unsupported	not found	no such
denied	invalid	timeout	timed out	abort
crash	crashed	panic	oops	bug
taint	call trace	segfault	killed	refused
reset	hang	corrupt	mismatch	deprecated
unknown	lacking	kaput		

2.4 Severity and status

Table 6 Severity and status

Level / status	Meaning
CRIT	Data loss, unbootable system, or active security exposure.
HIGH	Broken function or hardware at risk.
MED	Degraded function or performance with a visible impact.
LOW	Limited, conditional, or hygiene impact; action is optional or quick.
INFO	Explained and harmless; no action required.
OPEN	Present in the 2026-10-02 captures or awaiting an action; Sections 4–9 and 11.
CLOSED	Fixed and confirmed by the 2026-10-02 captures, answered, or a normal event; Section 10 only.
WATCH	Explained limit kept under observation; open, not counted as a finding.
BY DESIGN	Deliberate host choice surfacing in the logs; not counted.

3 System health

Table 7 System health checks

Check	Result	Evidence
ry-verify 7.219.0	PASS, exit 0	299 OK = 213 static + 86 runtime; 11 INFO notes (Table 14)
Fix script --verify	exit 0	7 steps ok; L-3 had no capture pair yet (FL 5)
Managed config, cmdline	17/17 checksums; 15/15 tokens live	No kernel parser rejections (VJ 377)
Kernel ring buffer	No taint, oops, or splat	1,359 lines; 13 keyword and 205 notice lines classified
Failed units	0 system, 0 user	FL 2
CPU vulnerabilities	None unmitigated	inxi: 14 not affected, 5 mitigated (BR 50–73)
GPU memory	VRAM 32,768 MiB	GTT 48,091 MiB, the kernel default (BR 1384)
amdgpu firmware	SMU initialized	DMUB 0x09005300; VCN ENC 1.24 / DEC 9; MES 0x00000092 (FL 2)
CPU / GPU policy	amd-pstate-epp, EPP performance; GPU DPM high	Microcode 0x0B700037 (early); governor powersave; dynamic_epp disabled (VJ 392)
zswap	Disabled (readback N)	Pool initialized by the vendor udev rule (I-1)
Temperatures at idle	CPU 49.6 °C, GPU 46.0 °C	Board 47.5 °C (BR 163); no fan RPM (I-13)
NVMe (Crucial P310 2 TB)	SMART PASSED, 42.9 °C	8.92 TB written, +0.40 TB in 5 days; 124 days powered on
Wi-Fi (MT7925)	Associated at 14.30 s	AP-advertised TX cap 30 dBm (watch)
Boot to root mount	7.22 s	3.87 s without log output first (Figure 3)

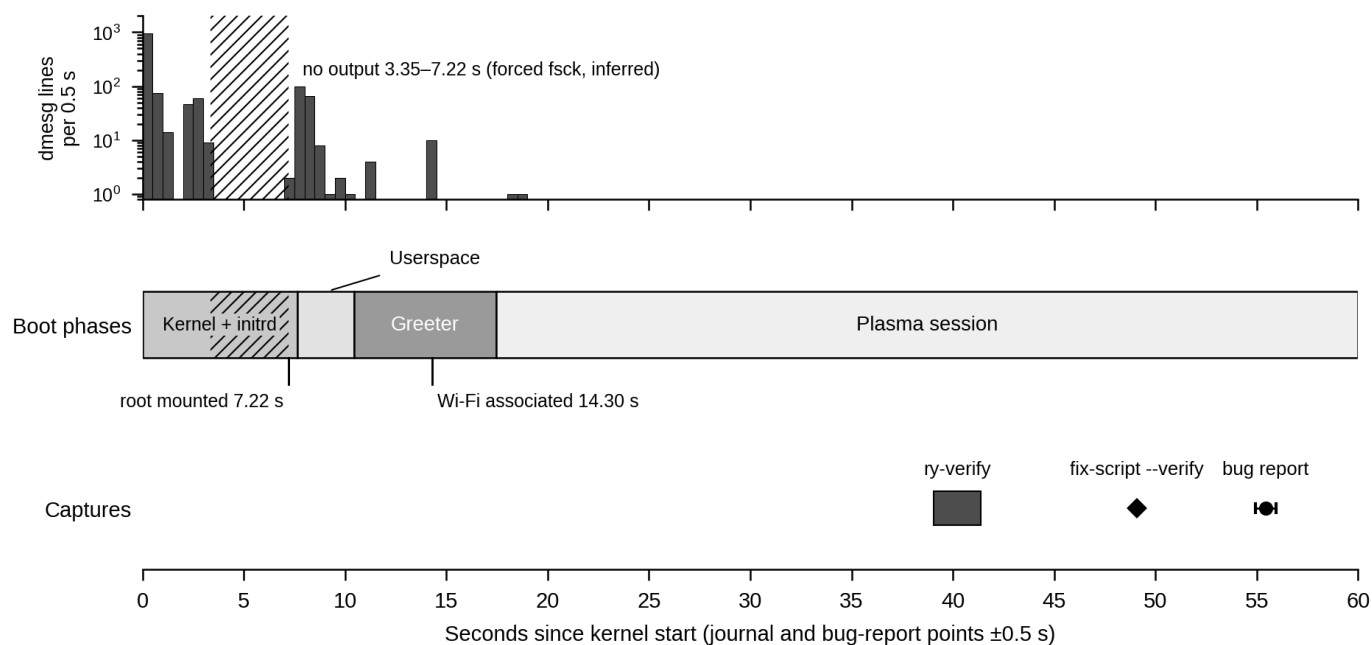


Figure 3 Current boot. Top: dmesg lines per 0.5 s (log scale); the empty stretch is the 3.87 s without output before the root mount (forced fsck in the initrd, inferred). Bottom: phase edges from dmesg (systemd in the initrd at 0.79 s, after switch-root at 7.65 s) and the journal (greeter ≈ 10.45 s, session ≈ 17.45 s); capture points from the JSONL, the fix-script log, and the bug report.

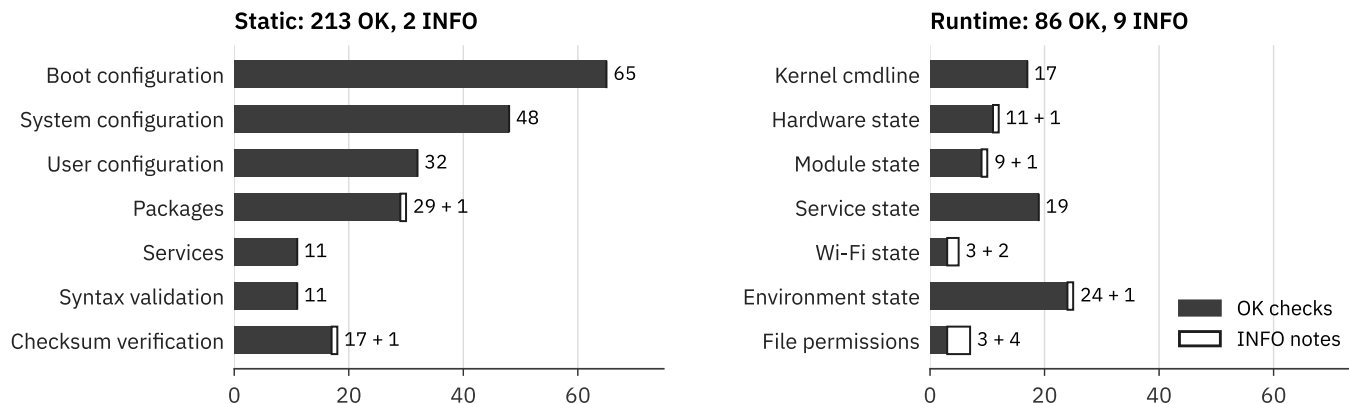


Figure 4 ry-verify 7.219.0 results by section. Static phase left, runtime right; dark bars are OK checks, outlined bars INFO notes (Table 14). Section counts exclude each phase's summary lines and match the 2026-09-27 run.

4 Open findings register

The register lists the open LOW finding first, then the INFO findings, which need no action. Line gives the first evidence line (BR and VJ prefixes: Appendix B); Page gives the finding card. Closed findings are in Section 10. Every row is OPEN.

Table 8 Open findings register

ID	Sev.	Finding	Area	Line	Action	Page
L-3	LOW	Logs carry identifiers the bug-report redactor misses	Privacy / log sharing	BR 8	Redact the Table 9 identifiers before posting	p. 9
I-1	INFO	zswap pool initialized although zswap.enabled=0	Memory / zswap	BR 1566	None	p. 11
I-2	INFO	bolt does not recognize the Strix Halo USB4 NHIs	USB4 / bolt	BR 1674	None; upstream bolt table entry	p. 11
I-3	INFO	amdgpu workqueue name truncated	Kernel / amdgpu DC	BR 1416	None; renamed in Linux 7.3-rc3	p. 11
I-4	INFO	MediaTek Bluetooth eSCO quirk notice	Bluetooth / btusb	BR 1635	None	p. 11
I-5	INFO	ACP70 finds no ASoC machine driver	Audio / ACP	BR 1598	None	p. 12
I-6	INFO	NVMe UUID and SGL notices	Storage / NVMe	BR 1335	None	p. 12
I-7	INFO	wpa_supplicant multicast RX registration unsupported	Network / wpa_supplicant	BR 1677	None	p. 12
I-8	INFO	TDX message on an AMD CPU	Kernel / TDX	BR 650	None	p. 12
I-9	INFO	KDE, portal, and D-Bus startup noise	Desktop / KDE, portal, D-Bus	BR 1676	None; skip the /usr and PAM workarounds	p. 13
I-10	INFO	plasmalogin-helper exits with status 255	Login / plasmalogin	BR 1721	None	p. 13
I-11	INFO	KWin warning at session hand-over	Desktop / KWin	BR 1684	None unless a login glitch appears	p. 13
I-12	INFO	Shutdown-only noise	Session / shutdown	BR 1808	None	p. 14
I-13	INFO	Platform visibility gaps (AER, fan RPM) and boilerplate	Platform / ACPI firmware	BR 164	None; note the AER blind spot	p. 14
I-14	INFO	inxi mislabels the CPU and GPU generation	Tooling / inxi	BR 35	None; note it when sharing the report	p. 14
I-15	INFO	EgisTec EH577 fingerprint reader has no driver	Input / fingerprint	BR 1395	None unless fingerprint login is wanted	p. 15
I-16	INFO	Secure Boot disabled	Boot / Secure Boot	BR 360	None; posture note	p. 15
I-17	INFO	Wireless-extensions warning from a Chromium-based process	Network / cfg80211	BR 1781	None	p. 15
I-18	INFO	KWin discards an unfinished kill prompt	Desktop / KWin	BR 1789	None unless it recurs	p. 15
I-19	INFO	Bluetooth audio device connects and disconnects	Bluetooth / BlueZ, pulseaudio-qt	BR 1782	None	p. 16
I-20	INFO	Task manager finds no desktop file for gldriverquery	Desktop / task manager	BR 1779	None	p. 16

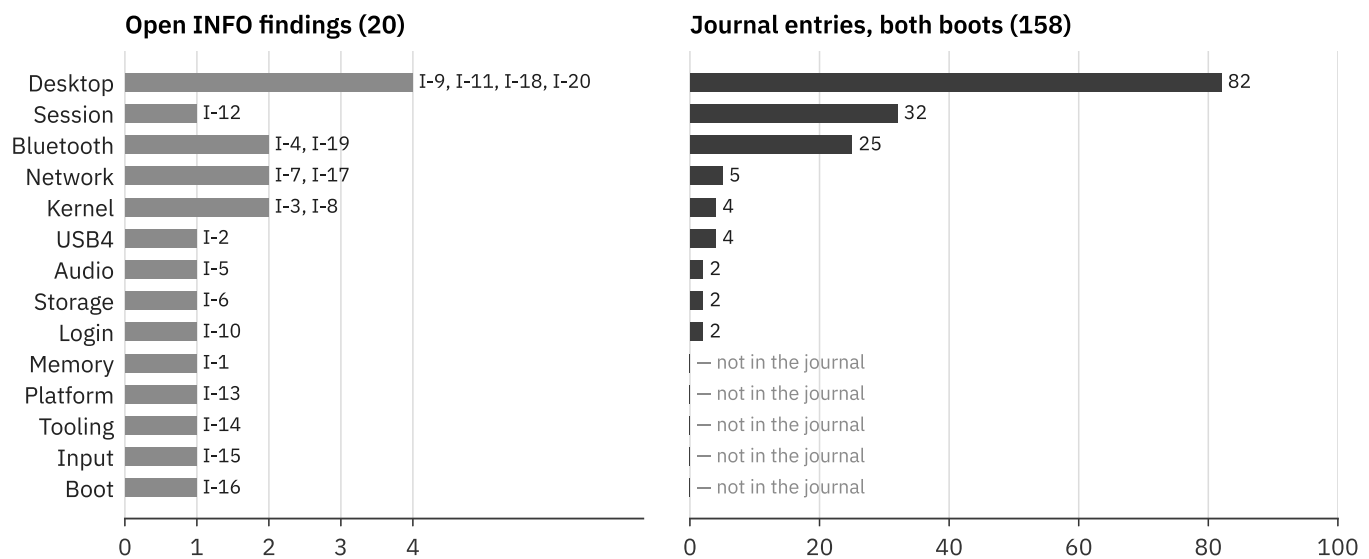


Figure 5 Open INFO findings by area (Table 8) against the journal entries they explain in both boots (Table 15). Desktop and shutdown noise make up 114 of the 158 entries; five areas surface only in dmesg or inxi.

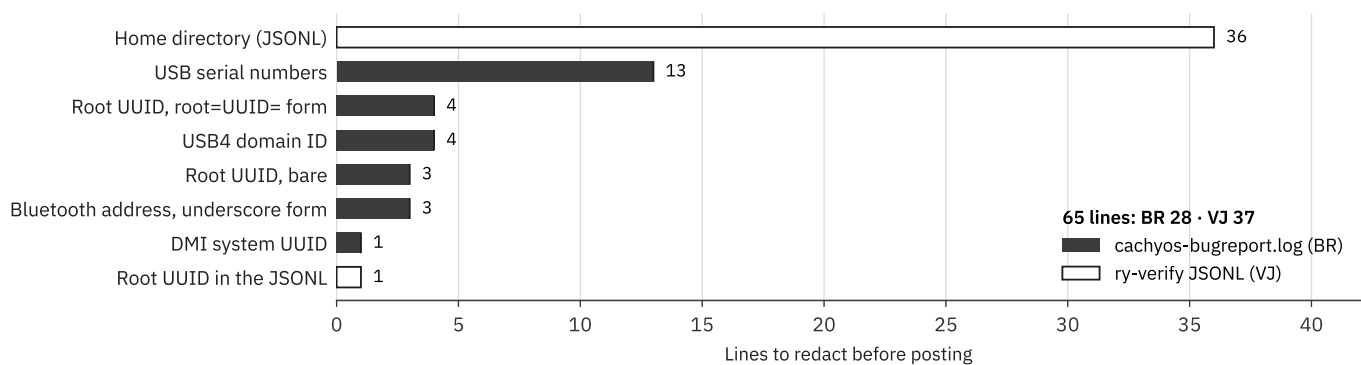
5 Open LOW finding

L - 3 Logs carry identifiers the bug-report redactor misses		LOW OPEN
LOW · OPEN · Privacy / log sharing · Commands: Section 11.1, p. 25 · Register, p. 8		
Evidence	cmdline root=UUID=[root UUID] (4 lines: header, inxi, dmesg x2) dmesg systemd[1]: Expecting device /dev/disk/by-uuid/[root UUID]... dmesg EXT4-fs (nvme0n1p2): mounted filesystem [root UUID] r/w (and re-mounted) inxi uuid: [DMI UUID] Firmware: UEFI vendor: American dmesg usb 3-2: SerialNumber: [serial] (Logitech USB Receiver; 3-3 EgisTec EH577, 3-4 POROSVOC) journal boltdd[837]: [<USB4 domain id>-domain0 ...] (4 lines) journal kded6[1313]: No object for name "bluez_output.[BT MAC].1" (3 lines) JSONL OK: root=UUID=[root UUID]: present (1 line) JSONL CHECK_FILE: /home/<user>/config/environment.d/... (36 lines)	
Lines	BR 8, 17, 28, 306, 516, 1285, 1310, 1346, 1399, 1405, 1459, 1469, 1561, 1674–1675, 1731–1732, 1787, 1791–1792 · VJ 66, 203, 347	
Cause	cachyos-bugreport.sh (CachyOS-Settings) redacts the hostname, user name, home directory, IPv4 addresses, colon-form MAC addresses, and email-shaped strings. It leaves the DMI system UUID, USB serial numbers, the root UUID, the USB4 domain ID, and underscore-form Bluetooth addresses (PipeWire node names). The JSONL embeds the root UUID and home paths.	
Impact	Only on sharing: the values tie posted reports to this machine and to a paired Bluetooth device.	
Fix	Post only copies with the Table 9 identifiers removed; Section 11.1 checks a copy.	
Status	Open for the 2026-10-02 pair: 28 bug-report and 37 JSONL lines carry identifiers (Table 9).	

Table 9 lists the identifier lines to remove before posting (Section 11.1).

Table 9 Identifier classes in the 2026-10-02 captures

Identifier class	Lines	Count
Root UUID, root=UUID= form (header, inxi, dmesg)	BR 8, 17, 306, 516	4
Root UUID, bare (systemd device, EXT4 mount and re-mount)	BR 1285, 1469, 1561	3
Root UUID in the JSONL	VJ 66	1
DMI system UUID (inxi)	BR 28	1
USB serial numbers (3 device-unique)	BR 1125, 1133, 1146, 1154, 1167, 1175, 1188, 1196, 1310, 1346, 1399, 1405, 1459	13
USB4 domain ID (bolt)	BR 1674–1675, 1731–1732	4
Bluetooth address with underscores (pulseaudio-qt)	BR 1787, 1791–1792	3
Home directory (JSONL)	VJ 203–205, 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 228–230, 232, 234, 236, 238, 240, 242, 244, 246, 248, 250, 252, 254, 256, 258, 260, 262, 264, 266, 346–347	36
Identifier lines in total	BR 28 · VJ 37	65

**Figure 6** Identifier lines to redact before posting, by class (Table 9). Home-directory paths in the JSONL account for 36 of the 65 lines.

6 Open INFO findings

No action is needed. I-1 to I-16 recur from 2026-09-27 with the same cause; I-17 to I-20 are new, from the longer previous boot.

I - 1 zswap pool initialized although zswap.enabled=0		INFO	OPEN
INFO · OPEN · Memory / zswap · Register, p. 8			
Evidence	[7.991631] zswap: loaded using pool zstd ry-verify: OK: zswap.enabled: N		
Lines	BR 1566 · VJ 398		
Explanation	CachyOS's 30-zram.rules, which also sets vm.swappiness to 150, writes N to zswap's enabled parameter when zram0 initializes. In Linux v7.2 that runtime write, with zswap still uninitialized because of zswap.enabled=0, runs zswap_setup(), which prints this line, and then stores N. zswap stays off (ry-verify readback N; inxi: zswap no), as ry-install intends with zram as the swap path; the only cost is the pool allocation.		
Action	None		

I - 2 bolt does not recognize the Strix Halo USB4 NHIs		INFO	OPEN
INFO · OPEN · USB4 / bolt · Register, p. 8			
Evidence	18:19:47 boltld[837]: [<USB4 domain id>-domain0] udev: failed to determine if uid is stable: unknown NHI PCI id '0x158d' 18:19:47 boltld[837]: [<USB4 domain id>-domain1] udev: failed to determine if uid is stable: unknown NHI PCI id '0x158e'		
Lines	BR 1674–1675		
Explanation	bolt keeps a table of known USB4/Thunderbolt native host interface (NHI) IDs to decide whether a host UUID survives reboots. Its table (0.9.11) lacks these Strix Halo IDs, so it treats the UUID as unstable — the safe default, as native USB4 host UUIDs can change at every boot. There is no effect without USB4 or Thunderbolt peripherals that need bolt authorization.		
Action	None; upstream bolt table entry		

I - 3 amdgpu workqueue name truncated		INFO	OPEN
INFO · OPEN · Kernel / amdgpu DC · Register, p. 8			
Evidence	[2.783176] workqueue: name exceeds WQ_NAME_LEN. Truncating to: hdmi_frl_status_polling_workque		
Lines	BR 1416		
Explanation	In v7.2, amdgpu_dm names a workqueue hdmi_frl_status_polling_workqueue, longer than the workqueue name limit, so the kernel truncates it once. Linux 7.3-rc3 shortens the name to hdmi_frl_status_polling_wq.		
Action	None; renamed in Linux 7.3-rc3		

I - 4 MediaTek Bluetooth eSCO quirk notice		INFO	OPEN
INFO · OPEN · Bluetooth / btusb · Register, p. 8			
Evidence	[8.455021] Bluetooth: hci0: HCI Enhanced Setup Synchronous Connection command is advertised, but not supported.		
Lines	BR 1635		
Explanation	btusb marks the Enhanced Setup Synchronous Connection command broken on every MediaTek controller, and the Bluetooth core lists active quirks at setup. Voice links use the legacy synchronous connection-oriented (SCO) setup instead; A2DP audio, which the SR-C20A soundbar uses, is unaffected.		
Action	None		

I - 5 ACP70 finds no ASoC machine driver		INFO	OPEN
INFO · OPEN · Audio / ACP · Register, p. 8			
Evidence	[8.106935] platform acp_asoc_acp70.0: warning: No matching ASoC machine driver found		
Lines	BR 1598		
Explanation	The Audio Co-Processor (ACP) driver logs this when no ACP machine description (DMIC or I2S codec) matches the board. Audio runs through the HDA codec (ALC897) and USB devices.		
Action	None		

I - 6 NVMe UUID and SGL notices		INFO	OPEN
INFO · OPEN · Storage / NVMe · Register, p. 8			
Evidence	[0.923433] nvme nvme0: passthrough uses implicit buffer lengths [18.614978] block nvme0n1: No UUID available providing old NGUID		
Lines	BR 1335, 1663		
Explanation	The first is an informational notice for a controller without scatter-gather list (SGL) support. The second is a one-time warning when userspace reads the namespace uuid attribute and the drive exposes only a namespace globally unique identifier (NGUID).		
Action	None		

I - 7 wpa_supplicant multicast RX registration unsupported		INFO	OPEN
INFO · OPEN · Network / wpa_supplicant · Register, p. 8			
Evidence	18:19:48 wpa_supplicant[930]: wlan0: nl80211: kernel reports: multicast RX registrations are not supported 18:19:48 wpa_supplicant[930]: p2p-dev-wlan0: nl80211: kernel reports: multicast RX registrations are not supported		
Lines	BR 1677–1678		
Explanation	nl80211 rejects multicast management-frame registrations with EOPNOTSUPP when the driver does not advertise NL80211_EXT_FEATURE_MULTICAST_REGISTRATIONS; wpa_supplicant logs it and continues. The P2P line remains because wpa_supplicant still creates that device, which NetworkManager no longer manages (L-6, closed).		
Action	None		

I - 8 TDX message on an AMD CPU		INFO	OPEN
INFO · OPEN · Kernel / TDX · Register, p. 8			
Evidence	[0.282216] virt/tdx: TDX not supported by the host platform		
Lines	BR 650		
Explanation	The kernel's Intel Trust Domain Extensions (TDX) host code logs this at error level whenever the CPU lacks TDX host support — always the case on AMD.		
Action	None		

I - 9 KDE, portal, and D-Bus startup noise		INFO	OPEN
INFO · OPEN · Desktop / KDE, portal, D-Bus · Register, p. 8			
Evidence	18:19:48 dbus-broker-launch[833]: Activation request for 'org.freedesktop.home1' failed: The systemd unit 'dbus-org.freedesktop.home1.service' could not be found. 18:19:48 dbus-broker-launch[957]: Service file '/usr/share/dbus-1/services/org.kde.dolphin.FileManager1.service' is not named after the D-Bus name 'org.freedesktop.FileManager1'. 18:19:56 ksmserver[1302]: Failed to register with host portal QDBusError("org.freedesktop.portal.Error.Failed", "Could not register app ID: App info not found for 'org.kde.ksmserver'") 18:19:56 org_kde_powerdevil[1346]: org.kde.powerdevil.chargethresholdhelper.getthreshold failed "Charge thresholds are not supported by the kernel for this hardware" 18:19:56 kded6[1304]: Failed enumerating MM objects: "org.freedesktop.DBus.Error.NameHasNoOwner" "Could not activate remote peer 'org.freedesktop.ModemManager1': activation request failed: unknown unit"		
Lines	BR 1676, 1679, 1690, 1693, 1701		
Explanation	- Portal host-registration failures (10 programs) are common on Plasma and had no visible effect; the circulating workaround adds placeholder .desktop files under /usr/share/applications, which the no-edits-under-/usr rule excludes. - dbus-broker flags three legacy service-file names; the greeter and the session each run a broker, so each notice appears twice per boot. The home1 request comes from pam_systemd_home while systemd-homed is unused; silencing it means editing a pambase-owned file. - ModemManager is not installed, so kded6 cannot enumerate modem objects. - powerdevil probes charge thresholds because the Logitech PRO X 2 exposes a HID++ battery (inxi: hidpp_battery_0), finds no kernel backlight (DDC/CI is off by design), and cannot check the screen configuration for button handling. - Also benign: QML override and deprecated-signal warnings, the gtk.portal fallback for Lockdown, a UPower owner race at greeter start, @DEFAULT_SOURCE@ lookups before WirePlumber picks a default source, and a screencast request for a closed window (18:18:48, previous boot).		
Action	None; skip the /usr and PAM workarounds		

I - 10 plasmalogin-helper exits with status 255		INFO	OPEN
INFO · OPEN · Login / plasmalogin · Register, p. 8			
Evidence	18:20:01 plasmalogin[920]: Auth: plasmalogin-helper exited with 255 [prev] 16:09:05 plasmalogin[926]: Auth: plasmalogin-helper exited with 255		
Lines	BR 1721, 1778		
Explanation	The greeter's authentication helper exits with 255 when the greeter stops after a successful hand-over to the user session; the same message is widely reported on working systems.		
Action	None		

I - 11 KWin warning at session hand-over		INFO	OPEN
INFO · OPEN · Desktop / KWin · Register, p. 8			
Evidence	18:19:55 kwin_wayland[959]: atomic commit failed: Permission denied [prev] 16:09:00 kwin_wayland[965]: atomic commit failed: Permission denied		
Lines	BR 1684, 1741		
Explanation	The permission error accompanies each greeter-to-session switch, consistent with the outgoing compositor losing DRM master. KDE bugs 521568 and 524540 report the same line at this transition, on NVIDIA systems with visible symptoms; here the session starts normally. The 2026-09-27 framebuffer warning does not recur.		
Action	None unless a login glitch appears		

I - 12 Shutdown-only noise		INFO	OPEN
INFO · OPEN · Session / shutdown · Register, p. 8			
Evidence	<pre>[prev] 18:19:28 plasmalogin[926]: Authentication error: PLASMALOGIN::Auth::ERROR_INTERNAL "Process crashed" [prev] 18:19:28 plasmalogin[926]: Auth: plasmalogin-helper (--socket /tmp/plasmalogin-auth-[id] --id 4 --start /usr/bin/startplasma-login-wayland --user plasmalogin --greeter) crashed (signal 15) [prev] 18:19:28 systemd[1]: sys-devices-virtual-misc-rfkill.device: Failed to enqueue SYSTEMD_WANTS job, ignoring: Transaction for systemd-rfkill.socket/start is destructive (systemd-reboot.service has 'start' job queued, but 'stop' is included in transaction). [prev] 18:19:28 systemd[1123]: sys-devices-pci0000:00-0000:00:08.1-0000:c6:00.6-sound-card0-controlC0.device: Failed to enqueue SYSTEMD_USER_WANTS job, ignoring: Transaction for sound.target/start is destructive (exit.target has 'start' job queued, but 'stop' is included in transaction). [prev] 18:19:28 plasmashell[1325]: PipeWire remote error: -2 target not found [prev] 18:19:28 kded6[1313]: context kaput [prev] 18:19:29 NetworkManager[900]: <warn> [1790990369.0699] dispatcher: (10) failed (after 0.001 sec): Could not activate remote peer 'org.freedesktop.nm_dispatcher': activation request failed: unit is invalid [prev] 18:19:29 systemd[1]: <email-address-redacted>: Failed with result 'exit-code'.</pre>		
Lines	BR 1808, 1811, 1817–1819, 1824, 1835, 1839		
Explanation	All occur during the 18:19:28 reboot. The session and greeter helpers, ended by SIGHUP and SIGTERM, are reported as crashes (signals 1 and 15). udev events cannot add jobs to the shutdown transaction of the system or the user manager. D-Bus activations (bluez, obex, UDisks2, NetworkManager dispatcher) are refused while units stop, which NetworkManager logs as a failed dispatcher call. PipeWire clients lose their server: Plasma and KWin log remote errors, and pulseaudio-qt logs “context kaput” and lookups of vanished nodes. One system unit whose name contains @ exits with failure as it stops; the redactor replaced the name (Section 9).		
Action	None		

I - 13 Platform visibility gaps (AER, fan RPM) and boilerplate		INFO	OPEN
INFO · OPEN · Platform / ACPI firmware · Register, p. 8			
Evidence	<pre>inxi: Fan Speeds (rpm): N/A [0.307147] acpi PNP0A08:00: _OSC: platform does not support [SHPCHotplug AER LTR DPC] [0.335423] acpi PNP0C02:01: Could not reserve [mem 0xfec00000-0xfec0ffff] [0.371843] usb usb2: We don't know the algorithms for LPM for this host, disabling LPM. [0.779023] usb 5-1: No LPM exit latency info found, disabling LPM.</pre>		
Lines	BR 164, 695, 941, 1128, 1270		
Explanation	Firmware keeps Advanced Error Reporting (AER), Latency Tolerance Reporting (LTR), and Downstream Port Containment (DPC) instead of granting them to the OS, and the BIOS has no option that hands them over, so the kernel's AER driver cannot report PCIe link errors — a clean ring is not proof of a clean link. No fan tachometer is exposed. The USB link power management (LPM) lines — four host controllers, and usb 5-1, the USB-C display cable, without exit-latency data — and the four reservation overlaps are standard; the ACPI core treats such overlaps as usually harmless.		
Action	None; note the AER blind spot		

I - 14 inxi mislabels the CPU and GPU generation		INFO	OPEN
INFO · OPEN · Tooling / inxi · Register, p. 8			
Evidence	<pre>inxi: bits: 64 type: MT MCP arch: Zen 6 note: 6 level: v4 note: check built: 2026+ inxi: process: TSMC n2/n3 (2,3nm) family: 0x1A (26) model-id: 0x70 (112) inxi: Radeon 8050S 8060S Graphics] driver: amdgpu v: kernel arch: RDNA-3 inxi: code: Phoenix process: TSMC n4 (4nm) built: 2023+ pcie: gen: 4</pre>		
Lines	BR 35–36, 76–77		
Explanation	inxi 3.3.41 maps AMD family 0x1A models from 0x50 up to Zen 6, so this model 0x70 CPU reads as Zen 6, and it labels the GPU Phoenix with RDNA-3. The Ryzen AI Max+ 395 is Zen 5 (16C/32T) with a gfx1151 Radeon 8060S, RDNA 3.5 (Strix Halo).		
Action	None; note it when sharing the report		

I - 15 EgisTec EH577 fingerprint reader has no driver

INFO

OPEN

INFO · OPEN · Input / fingerprint · Register, p. 8

Evidence [2.343811] usb 3-3: New USB device found, idVendor=1c7a, idProduct=0577, bcdDevice=10.41
[2.347684] usb 3-3: Product: EgisTec EH577

Lines BR 1395, 1397

Explanation The sensor (1c7a:0577) enumerates, but no driver binds: libfprint lists it as unsupported, and the upstream support request was closed in 2023. The only working path in circulation, the eh577-libfprint fork, runs EgisTec's Windows matching engine and replaces the packaged libfprint, which does not suit a managed host.

Action None unless fingerprint login is wanted

I - 16 Secure Boot disabled

INFO

OPEN

INFO · OPEN · Boot / Secure Boot · Register, p. 8

Evidence [0.002960] Secure boot disabled

Lines BR 360

Explanation UEFI Secure Boot is off, the firmware default; nothing in either capture depends on it.

Action None; posture note

I - 17 Wireless-extensions warning from a Chromium-based process

INFO

OPEN

INFO · OPEN · Network / cfg80211 · Register, p. 8

Evidence [prev] 16:09:20 kernel: warning: `ThreadPoolForeg' uses wireless extensions which will stop working for Wi-Fi 7 hardware; use nl80211

Lines BR 1781

Explanation Linux v7.2 prints this once per boot (net/wireless/wext-core.c) for the first process that uses the legacy wireless-extensions ioctls on a cfg80211 device; mt7925 sets WIPHY_FLAG_SUPPORTS_MLO, so the call is refused. ThreadPoolForeg is Chromium's ThreadPoolForegroundWorker thread cut to 15 characters; Chromium's network code uses SIOCGIWNAME and SIOCGIWESSID to classify an interface as Wi-Fi and read its SSID. Steam, whose web helper is Chromium-based, was starting four seconds earlier (I-20) and is the likely source. That process cannot read the SSID this way; connectivity is unaffected.

Action None

I - 18 KWin discards an unfinished kill prompt

INFO

OPEN

INFO · OPEN · Desktop / KWin · Register, p. 8

Evidence [prev] 16:27:40 kwin_wayland[1210]: QProcess: Destroyed while process ("/usr/lib/kwin_killer_helper") is still running.

Lines BR 1789

Explanation KWin starts kwin_killer_helper when a window it asked to close stops answering pings, to offer to terminate the application, and stops it when the window answers or goes away. Here the window went away while the helper still ran, so Qt warned as KWin discarded it: an application hung briefly on close at 16:27:40. The warning-level journal does not name it.

Action None unless it recurs

I - 19 Bluetooth audio device connects and disconnects		INFO	OPEN
INFO · OPEN · Bluetooth / BlueZ, pulseaudio-qt · Register, p. 8			
Evidence	<pre>[prev] 16:09:54 bluetoothd[840]: profiles/audio/a2dp.c:load_remote_sep() Unable to load LastUsed: rseid 2 not found [prev] 16:09:55 kded6[1313]: No object for name "bluez_output.[BT MAC].1" returning nullptr [prev] 16:57:32 bluetoothd[840]: src/profile.c:ext_io_disconnected() Unable to get io data for Hands-Free Voice gateway: getpeername: Transport endpoint is not connected (107) [prev] 16:57:32 kded6[1313]: No object for name "@DEFAULT_SINK@" returning nullptr</pre>		
Lines	BR 1782, 1787, 1790, 1793		
Explanation	At 16:09:54 bluetoothd found the device's cached LastUsed endpoint pair but not remote endpoint 2 among those the device now offered, so it skipped the cached choice (BlueZ a2dp.c); the device's audio node appeared a second later. At 16:57:32 the device closed its Hands-Free Profile (HFP) connection before bluetoothd read the socket's peer address (profile.c ext_io_disconnected; ENOTCONN, 107). Each node change makes pulseaudio-qt, used by Plasma's audio applet and kded6, look up nodes that are gone: the device's sink and source, the default sink and source, and the null sink. The device is not the SR-C20A that reconnect-soundbar manages.		
Action	None		

I - 20 Task manager finds no desktop file for gldriverquery		INFO	OPEN
INFO · OPEN · Desktop / task manager · Register, p. 8			
Evidence	<pre>[prev] 16:09:16 plasmashell[1325]: Failed to find service for Unity Launcher "gldriverquery.desktop" [prev] 16:09:16 plasmashell[1325]: Failed to find service for Unity Launcher "gldriverquery.desktop"</pre>		
Lines	BR 1779–1780		
Explanation	Plasma's task manager maps Unity LauncherEntry updates (badges, progress) to desktop files (plasma-desktop smartlauncherbackend.cpp). At 16:09:16 two updates named gldriverquery.desktop, which does not exist, so they were dropped. gldriverquery is the OpenGL driver-query helper Steam runs at start-up.		
Action	None		

7 Open watch and by-design items

Watch items are explained limits under observation: open, but not counted. By-design items are deliberate host choices and are not counted either. The normal reboot event is in Section 10.4.

Table 10 Watch items (open)

Item	Evidence	Watch for	Then
Wi-Fi TX power capped at 30 dBm	wlan0: Limiting TX power to 30 (30 - 0) dBm as advertised by the AP (BR 1661)	A lower cap after a router or firmware change, or Wi-Fi range or throughput complaints	Check the router's country and transmit-power settings; the cap follows the AP.
10 GbE ports down	enp193s0, enp197s0: Link is Down (BR 1645, 1647)	Latency or jitter in online games over Wi-Fi, or a cabled port that stays down	Connect one port to the router (tuning-audit item G-8) and confirm the link comes up.

Table 11 By-design items

Item	Evidence	Line	Why
IPv6 disabled	IPv6: Loaded, but administratively disabled	BR 1217	ipv6.disable=1
CPU idle limited to C1	ACPI: processor limited to max C-state 1	BR 1108	processor.max_cstate=1; firmware twin: Global C-state Control
Sleep states masked	ry-verify: the five sleep targets inactive	VJ 423	Masked by ry-install
swappiness 150, cache pressure 50, zram 93.93 GiB	inxi Swap block	BR 158	CachyOS-Settings: 30-zram.rules sets swappiness 150 when zram0 starts, 70-cachyos-settings.conf cache pressure 50; not in ry-install's overrides (VJ 160–176)
KWallet disabled	ksecretd: Lacking a socket, pipe: 0 env: 0	BR 1717	kdewallet off; see L-4, closed
Monitor brightness control off	no kernel backlight interface found; POWERDEVIL_NO_DDCUTIL=1	BR 1692	DDC/CI off by design (VJ 215)
32 GiB UMA carve-out; GTT at default	32768M of VRAM memory ready · 48091M of GTT memory ready	BR 1383	Permanent choice since 2026-09-06; the small-carve-out recipe is not used
fsck forced every boot	fsck.mode=force: active	VJ 362	fsck.mode=force; ≈3.9 s per boot, inferred
NetworkManager logging and checks	dispatcher LogLevelMax=notice · level=WARN · connectivity enabled=false	VJ 123, 134, 136	Fewer NM journal lines by design; no captive-portal detection

8 Coverage and double-check

Every stream was read in full; Tables 12–15 and Figures 7–9 show the classification.

Table 12 Coverage by stream

Stream	Size	Method	Result
inxi and scheduler blocks	291 lines	Read in full	Facts; feeds L-3, I-13, I-14, and the L-2 check
Kernel ring (dmesg)	1,359 lines	Read line by line	0 unexplained; 205 notice lines in 20 families (Table 13)
Kernel splat patterns	1,359 lines	Tainted:, Oops, WARNING: CPU, BUG:, Call Trace, general protection fault, kernel taint	0 matches
Journal, current boot	53 entries	Every entry attributed	53 / 53 (Table 15)
Journal, previous boot	114 entries	Every entry attributed	114 / 114 (Table 15)
Package list (cachyos-znver4)	41 packages	Read	Versions only
ry-verify JSONL	483 records	Every record read	299 OK; 11 INFO notes (Table 14); 0 FAIL, WARN, GEN_FAIL
Fix-script log	11 lines	Every record read	7 steps ok, L-3 none; exit 0
Keyword sweep, whole report	124 lines	38-term set (Table 5)	124 / 124: 109 journal, 13 dmesg, 2 report header

Table 13 dmesg notice lines by family

Notice family	Lines	First	Disposition
Firmware memory map, ACPI table reservations	80	BR 307	Boilerplate
PNP0C02 resource reservations	51	BR 905	I-13 for the four overlaps; the rest boilerplate
CPU vulnerability mitigations	10	BR 568	Posture: none unmitigated (Table 7)
USB link power management	5	BR 1128	I-13
xHCI quirk masks	4	BR 1117	Boilerplate
amdgpu optional features	7	BR 1375	Boilerplate: optional firmware absent; no runtime PM on an APU
systemd unmet conditions	18	BR 1292	Boilerplate: no measured boot, no hibernation
Audit subsystem off	3	BR 636	Boilerplate
Kernel command line	2	BR 306	By design: ry-install profile
Kernel and PCI notices	17	BR 334	Boilerplate
Secure Boot state	1	BR 360	I-16
CPU idle and IPv6 policy	2	BR 1108	By design (Table 11)
Previous reset reason	1	BR 1219	Closed event (Table 19)
TDX probe	1	BR 650	I-8
Workqueue name truncated	1	BR 1416	I-3
ACP machine driver	1	BR 1598	I-5
USB mic volume range	1	BR 1628	L-5 (closed)
Bluetooth eSCO quirk	1	BR 1635	I-4
10 GbE links down	2	BR 1645	Watch (Table 10)
Wi-Fi TX power cap	1	BR 1661	Watch (Table 10)

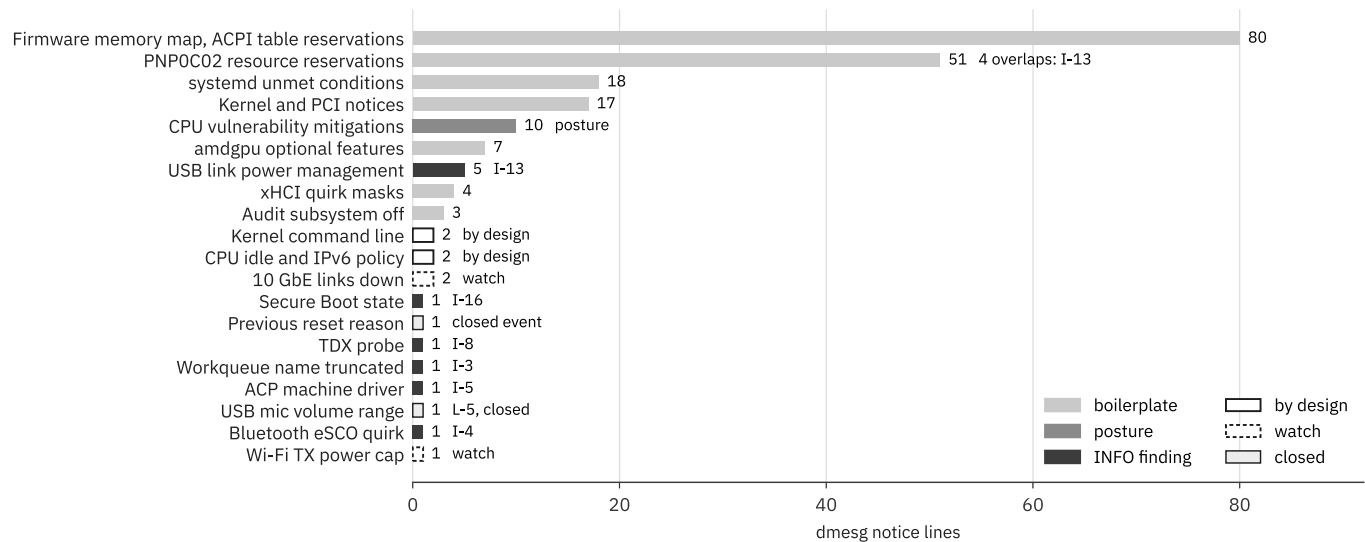


Figure 7 dmesg notice lines by family (Table 13), shaded by disposition. Boilerplate dominates; every finding-linked family is one to five lines.

Table 14 ry-verify INFO notes

VJ	Note	Reading
301	No IgnorePkg set	No package is held back from upgrades.
349	no .ry.bak copies (no run has rewritten a boot file or fstab, or they were removed by hand)	No pending backups of boot files or fstab.
382	Checking cpu0 (representative)	CPU policy checks read cpu0 as representative.
407	No module_blacklist= entry in KERNEL_PARAMS	No module blacklisted on the kernel command line.
432	NM Wi-Fi radio: enabled	Wi-Fi radio on.
434	firewall posture: ufw=inactive nft_rules=5	nftables ruleset active, input policy drop (VJ 195); ufw unused.
458	root filesystem: ext4	As expected.
470	/boot/loader/loader.conf: skipped (vfat — unix perms synthesized from mount options)	vfat has no Unix permissions; expected for /boot.
472	1 file(s) skipped on boot partition (vfat or undetermined fstype — unix perms not verifiable)	vfat has no Unix permissions; expected for /boot.
474	/boot/loader: skipped (vfat — unix perms synthesized from mount options)	vfat has no Unix permissions; expected for /boot.
476	1 dir(s) skipped on boot partition (vfat or undetermined fstype — unix perms not verifiable)	vfat has no Unix permissions; expected for /boot.

8.1 Journal attribution by finding

Shaded rows are closed findings whose lines still print as expected (L-4 accepted, L-5 kernel warning).

Table 15 Journal attribution by finding

ID	Finding	Status	Current boot	Previous boot
L-4	D-Bus-activated user units exit with failure after login	CLOSED	2	3
L-5	USB microphone hardware gain spans under 1 dB	CLOSED	2	2
L-6	NetworkManager warns on the Wi-Fi P2P device at every boot	CLOSED	0	0
I-2	bolt does not recognize the Strix Halo USB4 NHIs	OPEN	2	2
I-3	amdgpu workqueue name truncated	OPEN	1	1
I-4	MediaTek Bluetooth eSCO quirk notice	OPEN	1	1
I-5	ACP70 finds no ASoC machine driver	OPEN	1	1
I-6	NVMe UUID and SGL notices	OPEN	1	1
I-7	wpa_supplicant multicast RX registration unsupported	OPEN	2	2
I-8	TDX message on an AMD CPU	OPEN	1	1
I-9	KDE, portal, and D-Bus startup noise	OPEN	38	39
I-10	plasmalogin-helper exits with status 255	OPEN	1	1
I-11	KWin warning at session hand-over	OPEN	1	1
I-12	Shutdown-only noise	OPEN	0	32
I-17	Wireless-extensions warning from a Chromium-based process	OPEN	0	1
I-18	KWin discards an unfinished kill prompt	OPEN	0	1
I-19	Bluetooth audio device connects and disconnects	OPEN	0	23
I-20	Task manager finds no desktop file for gldriverquery	OPEN	0	2
Total	All journal entries		53	114

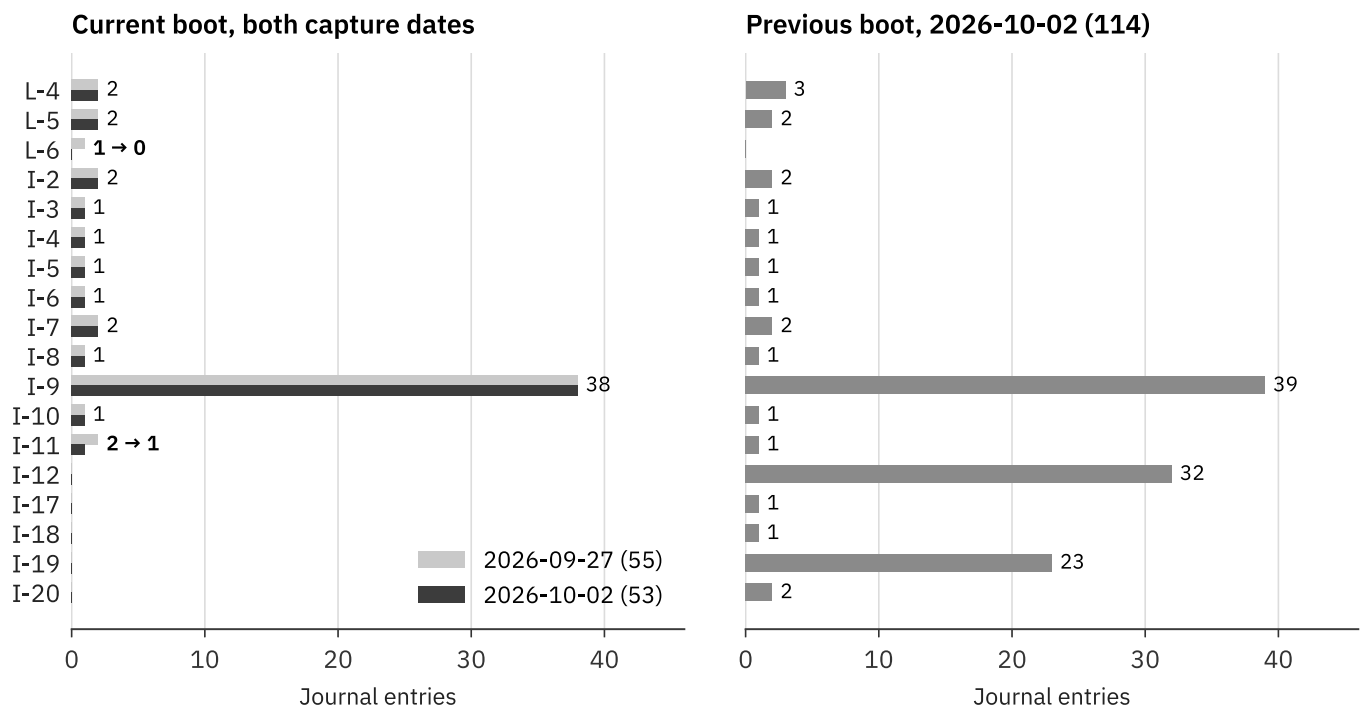


Figure 8 Journal entries per finding (Table 15). Left: current boot, 2026-09-27 (light) against 2026-10-02 (dark); L-6 drops to 0 and I-11 loses its framebuffer warning, and the rest recur at the same rate. Right: the 114 entries of the previous boot.

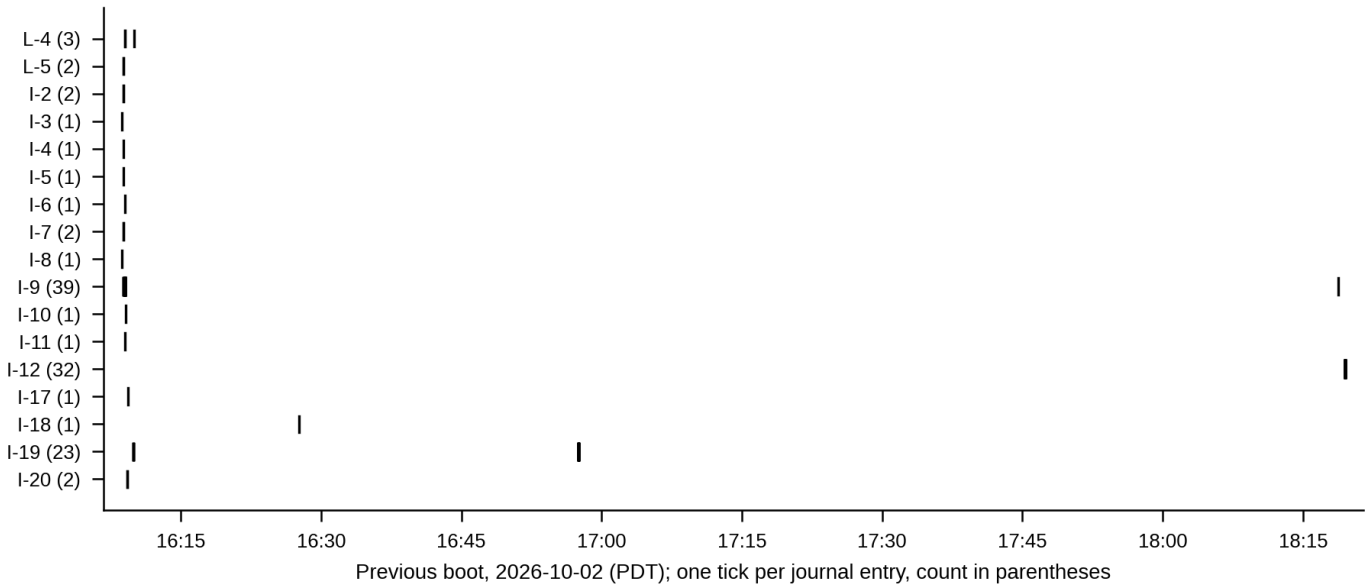


Figure 9 Previous boot, 16:08:42–18:19:29: journal entries by finding over time. Startup notices cluster at 16:08–16:10; I-18 is at 16:27:40, I-19 at 16:09:54 and 16:57:32, and I-12 is the 18:19:28 reboot.

9 Open unknowns and risks

What these captures cannot show, and how to close each item. The five unknowns of revision 19 that have since been answered are in Section 10.3.

Table 16 Open unknowns

Item	Why it matters	How to close
Idle capture	Both capture sets are from idle, ≈39–55 s after kernel start: no GPU reset, thermal, or frametime data under load.	Capture again after a gaming or LLM session; ry -dashboard --log records CPU and GPU temperature, clock, load, and power to CSV meanwhile.
Native AER unavailable	Firmware withholds AER (I-13), so PCIe link errors are not reported.	Account for it when diagnosing PCIe devices.
Redacted shutdown unit	A system unit with @ in its name failed as it stopped at 18:19:29 (I-12).	Section 11.3, first command, while the 18:19 boot is current.
Unnamed D-Bus failure, previous boot	The 16:10:00 failure (L-4) is unnamed; the fix script reads only the current boot.	Section 11.3, second command; expected: KSplash.
Window that hung on close	I-18 does not name the application.	Only if it recurs: note what was closed.
Bluetooth device	The I-19 device is not the SR-C20A; the capture shows only its address.	Section 11.3, third command.
Gap before the root mount	Grew from 3.06 s to 3.87 s; inferred to be the forced fsck, which neither capture times.	Section 11.3, fourth command.

10 Closed

Fixed and confirmed, answered, or normal; kept for the record. Nothing here needs action.

10.1 Closed findings register

Table 17 Closed findings register

ID	Finding	Closed	Evidence on 2026-10-02	Owner	Page
L-4	D-Bus-activated user units exit with failure after login	2026-09-30	FL 3: only the KWallet portal failed; 0 failed units	none needed	p. 22
L-1	Keychron Link receiver exposes a HID joystick interface	2026-10-02	FL 4: both pairs in the session (session=active)	ry-install 7.224.0	p. 23
L-5	USB microphone hardware gain spans under 1 dB	2026-09-30	FL 6: soft_mixer=true	ry-install 7.224.0	p. 23
L-6	NetworkManager warns on the Wi-Fi P2P device at every boot	2026-10-01	FL 7: unmanaged, 0 warnings; no warning in either boot	ry-install 7.224.0	p. 23
L-2	JACK clients bypassed PipeWire (jack2 provided libjack)	2026-10-02	FL 8: pipewire-jack; inxi lists pw-jack, no JACK server	ry-install 7.224.0	p. 24

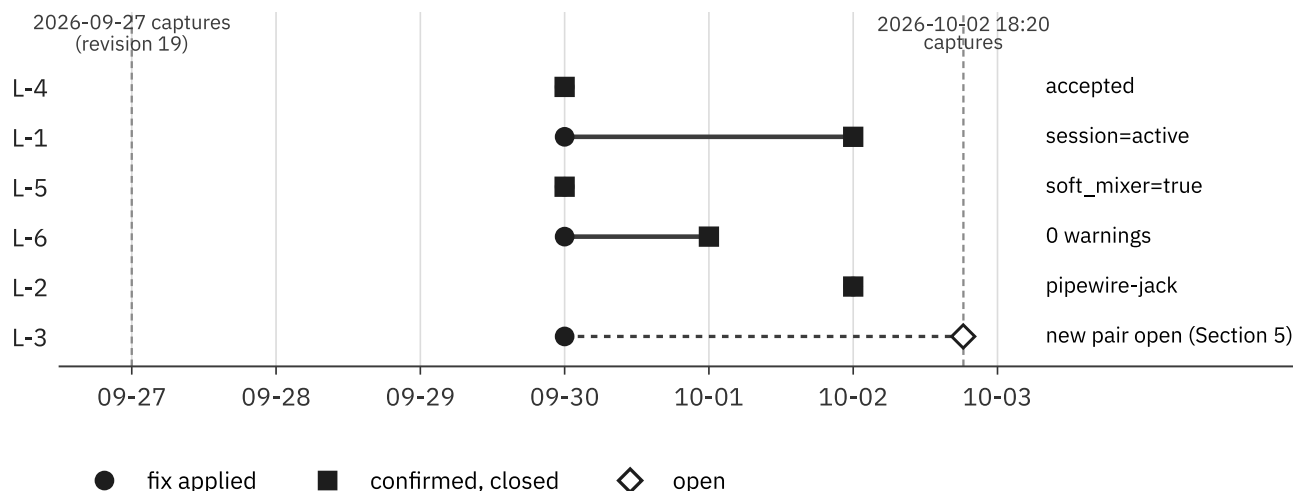


Figure 10 Fix history from Tables 17 and 20: when each fix was applied and when the captures confirmed it. L-3's earlier copies covered the 2026-09-27 pair; the 2026-10-02 pair is open.

10.2 Closed finding cards

L - 4 D-Bus-activated user units exit with failure after login		LOW	CLOSED
LOW · CLOSED 2026-09-30 · Session / systemd --user · Register, p. 22			
Evidence	<pre>fix log: fix=L-4 state=ok failed=0 journal_failures=1 units=dbus-:1.2-org.freedesktop.impl.portal.desktop.kwallet@0.service other= 18:19:56 ksecret[1572]: Lacking a socket, pipe: 0 env: 0 18:19:56 systemd[1116]: dbus-:<email-address-redacted>: Failed with result 'exit-code'. [prev] 16:10:00 systemd[1123]: dbus-:<email-address-redacted>: Failed with result 'exit-code'.</pre>		
Lines	BR 1717–1718, 1788 · FL 3		
Was	D-Bus-activated user units exited with failure after login; the bug report's email filter hid their names.		
Fix in place	None needed: kdwallet is disabled by design, so the KWallet portal and KSplash fail on start; the script reads their names from the user journal. Accepted 2026-09-30.		
Confirmed	One failure this boot, the KWallet portal, and no failed unit left (FL 3). The previous boot shows the pair at 16:09:01 and 16:10:00; Section 9 checks the second name.		

L - 1 Keychron Link receiver exposes a HID joystick interface		LOW	CLOSED
LOW · CLOSED 2026-10-02 · Input / HID · Register, p. 22			
Evidence	<pre>fix log: fix=L-1 state=ok pairs=0x3434/0x0e20,0x3434/0xd030 global=0x3434/0x0e20,0x3434/0xd030 file=ours session=active [0.900672] hid-generic 0003:3434:D030.0004: input,hiddev97,hidraw3: USB HID v1.11 Joystick [Keychron Keychron Link] on usb-0000:c8:00.0-1/input1</pre>		
Lines	BR 1328 · FL 4		
Was	The Keychron Link receiver (3434:d030) exposes interface 1 as a HID joystick that Proton and SDL games can take for a controller, which stops mouse input.		
Fix in place	~/.config/environment.d/60-sdl-ignore.conf sets SDL_GAMECONTROLLER_IGNORE_DEVICES=0x3434/0x0e20,0x3434/0xd030 for the session (2026-09-30).		
Confirmed	Both pairs reach the session (session=active, FL 4), so Steam inherits them. Steam Input must be off per game, and the proton-cachyos sdlinput and wayland modes need PROTON_NO_STEAMINPUT=0.		
Profile	ry-install 7.224.0 sets the pair in ~/.config/environment.d/10-environment.conf, and ry-verify checks it; delete 60-sdl-ignore.conf afterwards (O-3).		

L - 5 USB microphone hardware gain spans under 1 dB		LOW	CLOSED
LOW · CLOSED 2026-09-30 · Audio / USB mic · Register, p. 22			
Evidence	<pre>fix log: fix=L-5 state=ok soft_mixer=true [8.200254] usb 3-4: Warning! Unlikely small volume range (=100), linear volume or custom curve? [8.200257] usb 3-4: [12] FU [Mic Capture Volume] ch = 1, val = 0/100/1</pre>		
Lines	BR 1628–1629 · FL 6		
Was	The POROSVOC card (1d6b:a4a7) reports Mic Capture Volume as 0–100 in 1/256 dB, about 0.39 dB in total.		
Fix in place	WirePlumber rule 51-porosvoc-softmixer.conf sets api.alsa.soft-mixer = true for alsa_card.usb-POROSVOC* (2026-09-30).		
Confirmed	soft_mixer=true (FL 6). The kernel warning still prints twice per boot because the driver registers the hardware control; this is expected.		
Profile	ry-install 7.224.0 manages this path (mode 0600, its own text), and ry-verify reads the property through pactl (O-3).		

L - 6 NetworkManager warns on the Wi-Fi P2P device at every boot		LOW	CLOSED
LOW · CLOSED 2026-10-01 · Network / NetworkManager · Register, p. 22			
Evidence	<pre>fix log: fix=L-6 state=ok nm_state=unmanaged warnings=0</pre>		
Lines	FL 7		
Was	NetworkManager 1.54 set IPv4 forwarding on p2p-dev-wlan0, which has no kernel netdev, and logged ENOENT at every boot.		
Fix in place	/etc/NetworkManager/conf.d/90-no-p2p.conf unmanages type:wifi-p2p devices (2026-09-30); Wi-Fi Direct and Miracast are off.		
Confirmed	Unmanaged with 0 warnings (FL 7); no forwarding warning in either boot, against one per boot on 2026-09-27.		
Profile	ry-install 7.224.0 adds [device-no-p2p] to 99-cachyos-nm.conf, and ry-verify checks it; delete 90-no-p2p.conf afterwards (O-3).		

L - 2 JACK clients bypassed PipeWire (jack2 provided libjack)		LOW	CLOSED
LOW · CLOSED 2026-10-02 · Audio / JACK · Register, p. 22			
Evidence	fix log: fix=L-2 state=ok installed=pipewire-jack inxi: Server-2: PipeWire v: 1.6.9 status: n/a (root, process) with: inxi: 3: pipewire-alsa type: plugin 4: pw-jack type: plugin		
Lines	BR 119, 121 · FL 8		
Was	jack2 provided libjack, and inxi found a JACK server (jackd), so JACK clients bypassed PipeWire.		
Fix in place	pipewire-jack replaced jack2 1.9.22-2.1 on 2026-10-02 in a 14 s unattended upgrade (script 2.3.0).		
Confirmed	pipewire-jack installed (FL 8); inxi lists pw-jack under PipeWire and no JACK server.		
Profile	ry-install 7.224.0 declares pipewire-jack in PKGS_ADD; pacman's conflict keeps jack2 out (O-3).		

10.3 Closed unknowns

Table 18 Unknowns answered since revision 19

Unknown in revision 19	Answer	Evidence	Closed
Failed-unit state not captured	0 failed system units, 0 failed user units	FL 2	2026-10-02
MES firmware not logged	0x00000092, newer than the 0x91 reading of 2026-08-14	FL 2	2026-09-30
L-2 not confirmed by pacman	pacman listed jack2 1.9.22-2.1 alone; now pipewire-jack	FL 8	2026-10-01
Stray .ry.orig not visible	The root sweep of /etc finds none	FL 2, FL 9 (scope=root)	2026-10-02
L-1 session check pending	The drop-in reached the session	FL 4 (session=active)	2026-10-02

10.4 Closed events

Table 19 Normal events in the logs

Event	Evidence	Line	Status
Previous boot ended by reboot	software wrote 0x6 to reset control register 0xCF9	BR 1219	Closed; normal 18:19:28 reboot (shutdown lines: I-12)

10.5 Implementation record

When each step from revisions 1–19 was done, and with which script version. Step 4 covered the 2026-09-27 pair; L-3 is open for the new pair.

Table 20 Implementation record

Step	ID	Action	Done	Script	Result
1	BASE	Baseline checks	2026-09-30; root sweep 2026-10-02	2.0.x, 2.3.0	0 failed units, MES 0x00000092, no stray .ry.orig
2	L-4	Identify the failing D-Bus units	2026-09-30	2.0.1	KWallet portal and KSplash, accepted
3	L-1	Session-wide SDL drop-in	2026-09-30	2.0.x	session=active on 2026-10-02
4	L-3	-public copies of the 2026-09-27 pair	2026-09-30	2.0.x	Superseded; the 2026-10-02 pair is open, Section 5
5	L-5	WirePlumber soft-mixer rule	2026-09-30	2.0.x	soft_mixer=true
6	L-6	NetworkManager P2P drop-in	2026-09-30	2.0.x	0 warnings since the 2026-10-01 boot
7	L-2	pipewire-jack inside a full -Syu	2026-10-02	2.3.0	jack2 out in 14 s, third attempt
—	TIDY	Delete leftovers of earlier runs	2026-10-02	2.3.0	old=0, ry_orig=0

11 Open actions

Commands for the three open actions and the open unknowns, one per line. Section 11.5 is a checklist to tick as each expected result appears.

11.1 Redact the captures before posting (O-1)

Post only copies with the Table 9 identifiers removed, saved as bugreport-public.log and verify-public.jsonl. Check them from the directory that holds them:

MANUAL CROSS-CHECK

```
rg -c 'uuid: [[[:xdigit:]]{8}-|root=UUID=[[[:xdigit:]] bugreport-public.log verify-public.jsonl
rg -c 'SerialNumber: [^\[\s]]{8}-[[[:xdigit:]]{4}-domain' bugreport-public.log
rg -c '([[[:xdigit:]]{2}_){5}[[[:xdigit:]]{2}' bugreport-public.log verify-public.jsonl
rg -o -m 1 -r '$1' 'root=UUID=(\S{36})' cachyos-bugreport.log | rg -c -F -f - bugreport-public.log
rg -o -m 1 -r '$1' 'root=UUID=(\S{36})' cachyos-bugreport.log | rg -c -F -f - verify-public.jsonl
rg -c -F "$HOME" bugreport-public.log verify-public.jsonl
```

Expected: no output from any of them; rg exits 1 when nothing matches.

11.2 Merge the .pacnew files (O-2)

APPLY

```
sudo pacdiff
```

Expected: the seven files the 2026-10-02 upgrade listed; nothing to do if they were merged since.

11.3 Close the open unknowns

CHECK

```
journalctl -b -1 _PID=1 -p warning -o cat | rg 'Failed with result'
journalctl --user -b -1 -o cat -u 'dbus-*'
bluetoothctl devices
journalctl -b -o short-monotonic -u systemd-fsck-root.service
```

Expected, while the 18:19 boot is current (-b -1 is relative): the templated unit's name (I-12); the 16:10:00 D-Bus failure, likely KSplash; paired Bluetooth devices by name (I-19); and the root fsck's start and finish, whose difference measures the gap in Table 1.

11.4 Hand over to ry-install 7.224.0 (O-3)

After O-1 and O-2. ry-install 7.224.0 carries L-1 (ENV_VARS), L-2 (pipewire-jack in PKGS_ADD), L-5 (the same WirePlumber path, mode 0600), and L-6 ([device-no-p2p] in 99-cachyos-nm.conf); ry-verify checks all four.

HAND OVER

```
cd ~/ry-install
./ry-install.fish
rm ~/.config/environment.d/60-sdl-ignore.conf
sudo rm /etc/NetworkManager/conf.d/90-no-p2p.conf
./ry-verify.fish --verify
```

Expected: ry-install ends PASS or PASS-WITH-WARNINGS; after the next boot and login, ry-verify reports no FAIL. The fix script is retired: running it again would undo the profile.

11.5 Checklist

ID	Action	Done when	Date / initials
☐ O-1	Redact the 2026-10-02 captures (Section 11.1)	All six checks print nothing	
☐ O-2	Merge the .pacnew files (Section 11.2)	pacdiff lists nothing left to merge	
☐ O-3	Hand over to ry-install 7.224.0 (Section 11.4)	PASS; ry-verify shows no FAIL after a reboot	
☐ U	Close the open unknowns (Section 11.3)	Unit name, D-Bus unit, device name, fsck timing noted	

A Environment snapshot

Table A-1 Environment snapshot

Item	Value
Machine	Beelink GTR9 Pro (DMI: AZW GTR Pro); BIOS GTRPRPI1001C (2026-05-12); AGESA StrixHaloPI-FP11 1.0.0.1c
CPU	AMD Ryzen AI Max+ 395, 16C/32T; microcode 0x0B700037 (early update from 0x0B700034)
GPU	Radeon 8060S (1002:1586, gfx1151); VRAM 32,768 MiB; GTT 48,091 MiB
Memory	96 GiB outside the carve-out, 93.93 GiB usable; zram swap 93.93 GiB (zstd)
Storage	Crucial P310 2 TB (CT2000P310SSD8, firmware V8CR001); ext4 root (43.4% used), vfat /boot
Network	MediaTek MT7925 Wi-Fi 7 (mt7925e) and Bluetooth 5.4 (13d3:3604); 2 × Realtek RTL8127 10 GbE (r8169)
Audio	HDA ALC897; POROSVOC USB audio; AMD ACP (no machine driver); PipeWire 1.6.9 with pw-jack
Input	Keychron K2 HE (3434:0e20), Keychron Link receiver (3434:d030), Logitech Lightspeed receiver (046d:c54d) with a PRO X 2 mouse, EgisTec EH577 reader (no driver)
Display	BenQ PD3420Q, 3440 × 1440 at 60 Hz on DP-2 through an Anker USB-C to HDMI cable (usb 5-1)
Kernel	7.2.8-2-cachyos (clang 23.1.1); sched-ext disabled; fallback linux-cachyos-lts 6.18.52; both newer than 6.18.4, the toolboxes' validated kernel
Userspace	systemd 262 · KDE Plasma 6.7.5 (Wayland) · Mesa 26.2.4 · PipeWire 1.6.9 · inxi 3.3.41
Profile	ry-verify 7.219.0, profile gtr9_pro; gtr9-postboot-fix 2.3.0 at capture, retired since

B Abbreviations

Table B-1 Abbreviations and conventions

Term	Meaning	Term	Meaning
A2DP	Advanced Audio Distribution Profile (Bluetooth stereo audio)	MES	MicroEngine Scheduler, AMD GPU firmware that schedules queues
ACP	Audio Co-Processor, the AMD audio block	MLO	Multi-Link Operation (Wi-Fi 7)
ACPI	Advanced Configuration and Power Interface	NGUID	Namespace Globally Unique Identifier (NVMe)
AER	Advanced Error Reporting (PCI Express)	NHI	Native Host Interface, the USB4 host controller function
AGESA	AMD Generic Encapsulated Software Architecture, the platform firmware base	NM	NetworkManager
AP	Wi-Fi access point	NVMe	Non-Volatile Memory Express, the SSD interface
ASoC	ALSA System on Chip, the Linux audio layer for on-chip audio	O-n	Open action identifiers (Table 2)
BR	Line number in cachyos-bugreport.log	P2P	Peer-to-peer; here Wi-Fi Direct
DC	Display Core, the amdgpu display driver	PAM	Pluggable Authentication Modules
DDC/CI	Display Data Channel Command Interface (monitor control)	PCIe	PCI Express
DMI	Desktop Management Interface; the SMBIOS identity data	PDT	Pacific Daylight Time (UTC-7)
DMIC	Digital microphone	QML	Qt Modeling Language, used by KDE interfaces
DMUB	AMD display microcontroller firmware	RDNA	AMD GPU architecture family
DPC	Downstream Port Containment (PCI Express)	SCO	Synchronous Connection-Oriented link (Bluetooth voice)
DPM	Dynamic Power Management (amdgpu)	SDL	Simple DirectMedia Layer, the input and media library many games use
DRM	Direct Rendering Manager, the kernel graphics subsystem	SGL	Scatter-Gather List (NVMe data transfer)
EPP	Energy Performance Preference (amd-pstate)	SMART	Self-Monitoring, Analysis and Reporting Technology
eSCO	Enhanced Synchronous Connection-Oriented link (Bluetooth voice)	SMBIOS	System Management BIOS, the firmware identity tables
FL	Line number in the fix-script log	SMU	System Management Unit, AMD power-management firmware
GTT	Graphics Translation Table; system memory the GPU can map	SSID	Service Set Identifier, a Wi-Fi network name
HCI	Host Controller Interface (Bluetooth)	TDX	Intel Trust Domain Extensions
HDA	High Definition Audio	TX	Transmit
HFP	Hands-Free Profile (Bluetooth voice)	UEFI	Unified Extensible Firmware Interface
HID	Human Interface Device (USB input class)	UMA	Unified Memory Architecture; here the fixed VRAM carve-out
I2S	Inter-IC Sound, a serial audio bus	UUID	Universally Unique Identifier
JACK	JACK Audio Connection Kit, a low-latency audio server and API	VCN	Video Core Next, the AMD video codec engine
JSONL	JSON Lines: one JSON object per line	VJ	Line number in the ry-verify JSONL
L-n, I-n	LOW and INFO finding identifiers	VRAM	Video memory; here the UMA carve-out
LPM	Link Power Management (USB)	wext	Wireless Extensions, the legacy Wi-Fi ioctl interface
LTR	Latency Tolerance Reporting (PCI Express)		